

## Adding and subtracting time



- 1 Evaluate the following in hours and minutes:
  - a 11 hours – 43 minutes
  - b 11 hours 16 minutes – 5 hours
  - c 5 hours 10 minutes + 20 minutes
  - d 4 hours 10 minutes – 35 minutes
  - e 3 hours 3 minutes + 3 hours 29 minutes
  - f 3 hours 20 minutes – 2 hours 10 minutes
  - g 2 hours 50 minutes + 3 hours 40 minutes
  - h 6 hours 10 minutes – 2 hours 30 minutes
  - i 7 hours 59 minutes + 12 hours 11 minutes
  - j 10 hours 33 minutes – 7 hours 57 minutes
  - k 10 hours 43 minutes – 2 hours 29 minutes
- 2 How many hours and minutes left until midnight if the time is 16:50?
- 3 Find the following times, expressing your answer in 24-hour time:

|                                       |                              |
|---------------------------------------|------------------------------|
| a 4 hours after 16:28                 | b 7 hours after 22:53        |
| c 12 hours before 9:57                | d 13 hours before 10:43      |
| e 30 minutes after 9:30 am.           | f 30 minutes after 11:40 pm. |
| g 9:30 am plus half an hour.          | h 7:00 pm plus 24 minutes.   |
| i 45 minutes after 11:20 am.          | j 20 minutes after 3:50 pm.  |
| k 10:30 am plus two and a half hours. | l 9:10 pm plus 63 minutes.   |
- 4 Find the 24-hour time and day after evaluating the following:

|                                        |                                        |
|----------------------------------------|----------------------------------------|
| a Saturday 13:15 – 15 hours 18 minutes | b Thursday 13:20 + 21 hours 59 minutes |
|----------------------------------------|----------------------------------------|
- 5 James went to a movie at 11:50 am. The movie went for 1 hour and 10 minutes. What time will be it when the movie finishes? Write your answer using 12-hour time.
- 6 Sharon starts watching a movie at half past eleven am, which goes for 2 hours and 30 minutes. What time will be it when the movie finishes? Write your answer using 12-hour time.
- 7 Valentina starts work at 7 am and finishes work at 5:30 in the evening. How many hours and



- 8 A train ride between two cities takes 9 hours and 57 minutes. On a train ride between these cities, if you leave one city at 23:44, at what time will you arrive at the other city? Give your answer in 24-hour time.
- 9 In a relay of four runners, Karen ran for 1:46, Janet ran for 1:18, Tania ran for 1:16 and Tina ran for 1:42. How long was their total time for the relay?
- 10 A ship arrives at 17:28 at Melbourne. If the trip took 2 hours and 16 minutes, at what time did the ship depart from Sydney? Express your answer in 24-hour time.
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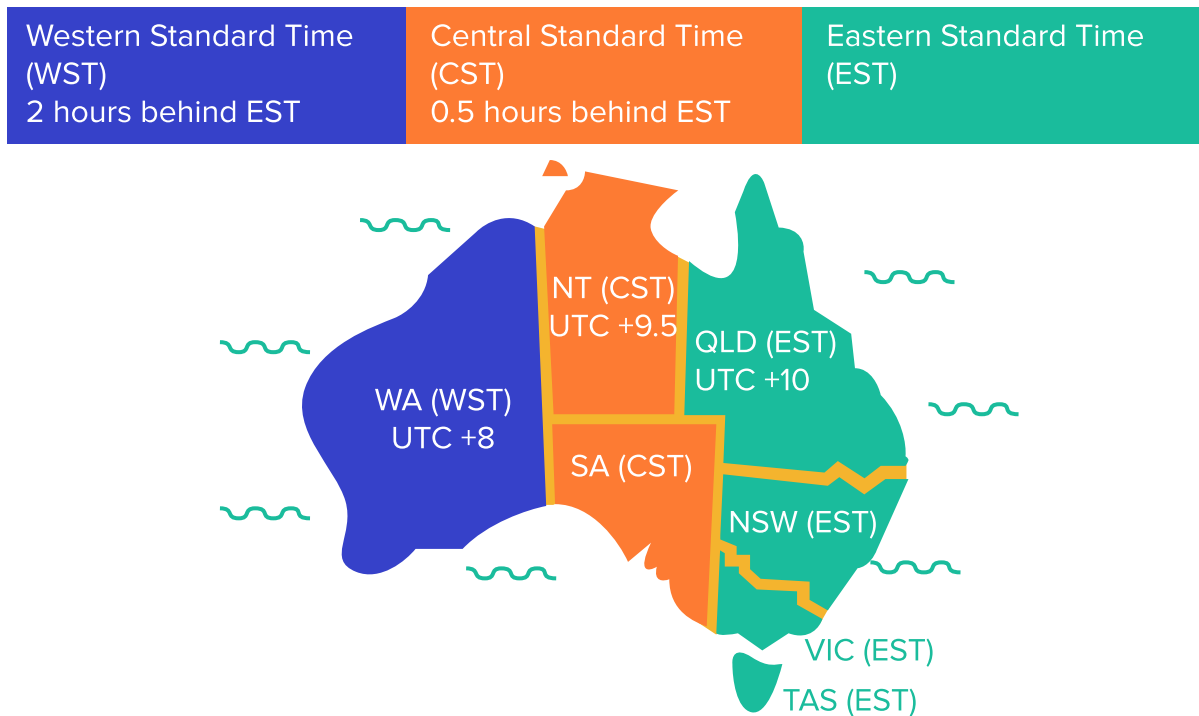
## Time between events

- 11 James leaves home at 2 pm and gets home at 5 pm. How many hours was James out?
- 12 Luke starts school at 9 am and finishes at 3 pm. How many hours is Luke at school each day?
- 13 Ellie's family leaves her home at 9:17 am to visit her grandmother in a different state. She arrives at her grandmother's house at 3:17 pm. How long did the trip take?
- 14 A song clip starts playing at 7:55 pm and finished at 8:11 pm. How long is the song clip in minutes?
- 15 A taxi departs at 13:32 from Sydney and arrives at 20:14 at Melbourne. How long did the taxi take for the trip?
- 16 A match of hockey begins at 7:31 and finishes at 8:55. If the half-time break goes for 15 minutes, what is the duration of one half of hockey?



## Time zones in Australia

17 Consider the map below:



- a Determine the time in Perth when it is 6:42 pm in Hobart.
- b Complete the table to show the equivalent time in each state:

| Western Australia (WA) | South Australia (SA) | Queensland (QLD) |
|------------------------|----------------------|------------------|
| 6:15                   |                      |                  |
|                        |                      | 11:45            |
|                        | 10:15                |                  |
| 5:25                   |                      |                  |

- c Complete the following table to show the equivalent time in each time zone:

| WST  | CST  | EST   |
|------|------|-------|
|      | 5:25 |       |
| 5:11 |      |       |
|      |      | 12:25 |
|      | 7:17 |       |

18 If it is 4:00 pm Daylight Savings Time (DST) in South Australia (SA), determine the time in:

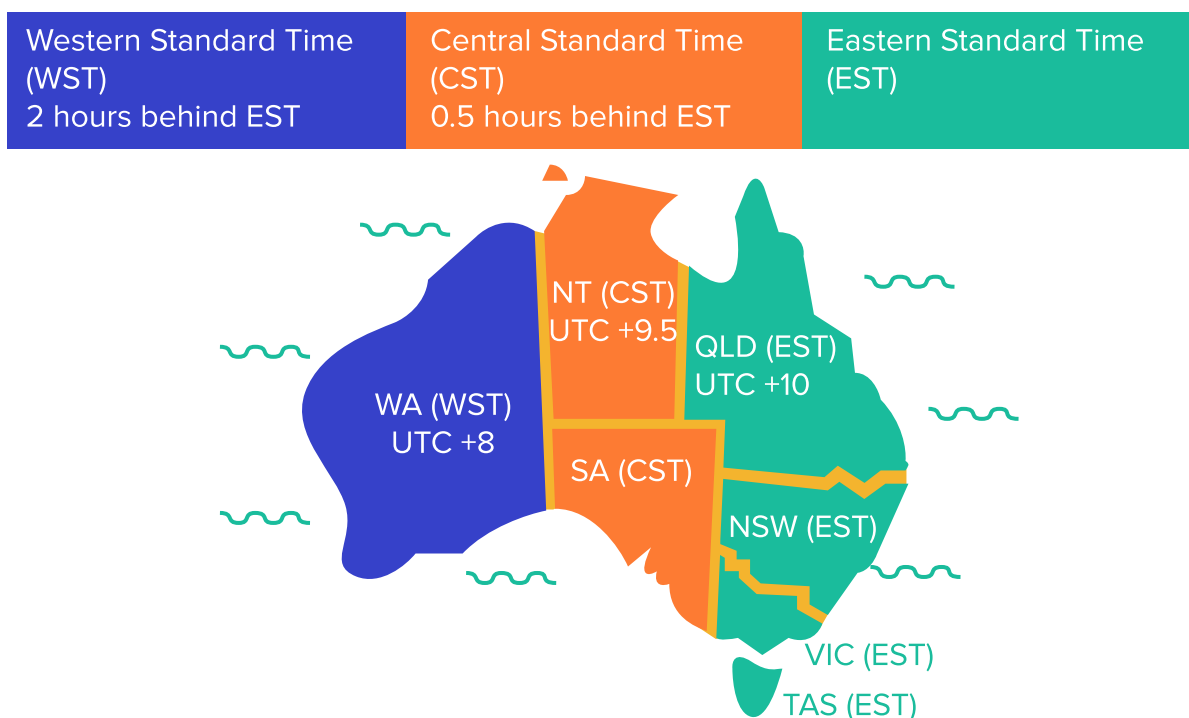
- a New South Wales (NSW).
- b Western Australia (WA).
- c Queensland.
- d Northern Territory (NT).

19 Australia is divided into 3 time zones:

- AEST (Eastern Standard Time) encompasses QLD, NSW, the ACT, VIC and TAS.
- ACST (Central Standard Time) encompasses the NT and SA, and is half an hour behind AEST.
- AWST (Western Standard Time) encompasses WA and is two hours behind AEST.

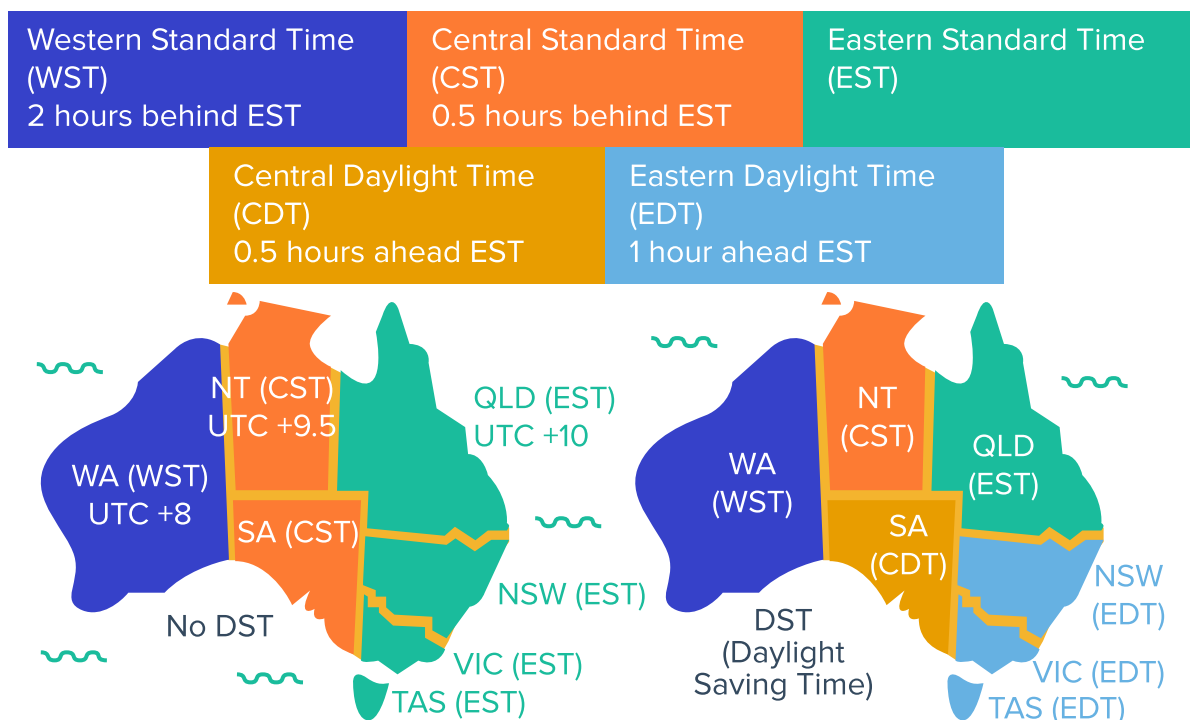
- a If it is 12:10 am AEST, determine the following, giving your answers in 24-hour time:
  - i The time in ACST.
  - ii The time in AWST.
- b If it is 1:16 pm in ACST, determine the time in AWST.

20 Han and Amelia board a train at 8:52 pm on Thursday from Sydney, New South Wales towards Adelaide, South Australia.



- a They arrived in Adelaide, South Australia on 6:48 pm Friday. Determine the duration of their train trip.
- b Han boards a train from Adelaide, South Australia at 8:43 pm on Friday to Darwin, Northern Territory. He arrives at 8:11 pm Saturday. Determine the duration of his trip.
- c Amelia boards a train from Adelaide, South Australia at 7:53 pm Friday towards Perth, Western Australia. She arrives at 10:54 am. Determine the duration of her train trip.

- 21** The map below shows the time zones across the Australian States in standard time and daylight saving time:



Calculate the flight times of the following trips:

- The airplane leaves Broome Airport in Western Australia at 7:30 pm and arrives at Albury (operating on daylight savings time) Airport in Victoria at 3:52 am.
  - The airplane leaves Albury Airport in Victoria at 7:14 pm and arrives at Adelaide Airport in South Australia (operating on daylight savings time) at 9:10 pm.
  - The airplane leaves Adelaide Airport in South Australia at 9:53 am and arrives at Broome Airport in Western Australia (operating in daylight savings time) at 3:41 pm.
- 22** If it is 16:14 in Sydney in Daylight Savings Time, determine the time it would be in Eastern Standard Time.
- 23** Determine the time it would be in Eastern Standard Time for the following:
- 2:49 am in Sydney in Daylight Savings Time.
  - 11:50 pm in Sydney in Daylight Savings Time.
- 24** If it is 14:35 in Sydney in Eastern Standard Time, determine the time it would be in Daylight Savings Time.
- 25** Determine the time it would be in Daylight Saving Time for the following:
- 9:45 am in Sydney in Eastern Standard Time.
  - 5:13 pm in Sydney in Eastern Standard Time.

**26** If it is 4 am in standard time, what time is it in daylight savings time (DST)?



**27** If it is 8 am in daylight savings time (DST), what time is it in standard time?